1 Internet Design Goals

Primary Goal

- Integrate multiple independently administered networks into a single global communication system

Secondary Goals

- **Survivability** — continue operation despite loss of links or routers
- **Multiple services** — support different applications (web, email, streaming, file transfer)
- **Multiple networks** — operate over different physical and link technologies
- **Distributed management** — no central control of resources
- **Cost effectiveness** — scalable and efficient use of infrastructure
- **Easy attachment** — low barrier for hosts to join the network
- **Accountability** — resources and usage must be measurable and attributable

2 Shared Medium and Multiplexing

Why Multiplexing

- Communication links are usually **shared** by many data flows
- The network must support:
  - **Multiplexing** — combining multiple flows on one link
  - **Demultiplexing** — separating flows at the receiver

Multiplexing Methods

- **Time Division Multiplexing (TDM)** — each flow gets time slots
- **Frequency Division Multiplexing (FDM)** — each flow gets a frequency band
- **Code Division Multiplexing (CDM)** — flows use different codes
- **Orthogonal Multiplexing** — resource slices do not interfere

Motivation for Packet Switching

- Fixed resource allocation can waste bandwidth for bursty traffic
- Packet switching enables **statistical multiplexing**

3 Circuit vs Packet Switching

Circuit Switching

- Dedicated route between source and destination
- Resources reserved for entire session
- Single continuous stream of bytes
- Uses TDM, FDM, or CDM
- Guaranteed performance
- Failure breaks the connection
- Example: Traditional telephone networks

Packet Switching

- Message is divided into packets
- Medium used only during packet transmission
- Statistical multiplexing
- Each packet is self-contained
- Packets routed independently
- Best-effort service
- Robust to failures via rerouting

4 Packet Switching: Key Properties (Exam Framing)

- Good for **bursty traffic**
- Supports **independent routing** per packet
- Enables **network sharing**
- **Cuts messages into packets**
- More **robust than circuit switching**
- Not originally designed for phone systems

5 Protocols

Definition

A protocol defines:

- Roles of communicating entities
- Format (syntax) of messages
- Order of message exchange
- Actions taken on transmission or receipt

Essential Parts of a Protocol

- **Syntax** — structure and format
- **Semantics** — meaning of fields and actions
- **Order of interactions** — sequencing of messages

Not Part of a Protocol

- Physical transmission medium
- Private authorization codes
- Implementation details

Client–Server Model

- **Client** initiates communication
- **Server** waits for requests
- Many Internet protocols use request–response patterns

6 Internet Protocol Stack

Layer	Protocols	Unit
Application	HTTP, FTP, SMTP, DNS	Message
Transport	TCP, UDP	Segment
Network	IP, ICMP, OSPF, BGP	Packet
Link	Ethernet, WiFi	Frame
Physical	802.11, 1000BASE-T	Bits

Layer Responsibilities

- **Application** — message meaning and rules
- **Transport** — process delivery, ports, reliability, flow control
- **Network** — addressing and routing between networks
- **Link** — delivery to next hop, framing, MAC addresses
- **Physical** — bit encoding for medium

Encapsulation (Sending)

Application → Transport (ports) → Network (IP) → Link (MAC, CRC) → Physical (bits)

At Routers

- Remove link header
- Inspect destination IP
- Forward using routing table
- Do NOT examine transport or application layers

7 IP Addresses

Basics

- Each interface has an IP address
- IPv4 uses 32 bits
- Written as a.b.c.d (each 0–255)

Routing Behavior

- Sender sends packet to router
- Router forwards based on destination IP
- Decision made using forwarding table

8 Autonomous Systems (AS)

- Internet divided into ASs
- Each AS owns an IP range
- Types: last-mile, backbone, consortium
- Connect at Internet Exchanges

Peering vs Transit

- **Peering** — exchange customer traffic
- **Transit** — one AS carries traffic to another AS

9 CIDR (Classless Inter-Domain Routing)

Notation

$$a.b.c.d/n$$

- First n bits = network
- Remaining = host

Formulas

$$\text{Host bits} = 32 - n$$
$$\text{Total addresses} = 2^{32-n}$$
$$\text{Usable hosts} = 2^{32-n} - 2$$

Netmask Conversion

- First n bits = 1, rest = 0
- Convert binary to decimal

Same Network Test

Bitwise AND both IPs with netmask. If results match, same network.

Reserved Addresses

- First = network address
- Last = broadcast address

10 CIDR Splitting and Aggregation

Splitting

- Fix the next host bit
- Increases prefix by 1
- Creates two equal subnets

Aggregation

- Combine prefixes only if no extra addresses are added
- One prefix may already cover another

11 Routing

Forwarding Table

Maps IP ranges to next hop.

Longest Prefix Match

Most specific prefix wins.

Inter-AS

- **BGP** — path vector protocol
- Uses AS paths to prevent loops

Intra-AS

- **OSPF** — link-state
- **RIP** — distance-vector

12 DHCP

DORA

1. Discover (broadcast)
2. Offer
3. Request
4. ACK

Server Provides

- IP address
- Netmask
- Default gateway
- Lease time
- DNS server

13 Traceroute

- Finds path using error messages
- Each hop returns RTT

Link Delay

$$\text{Link delay} = RTT(n) - RTT(n-1)$$

Notes

- Reverse route may differ
- Routers have multiple interfaces/IPs

14 Self-Contained Packets

- Must include:
  - Source IP
  - Destination IP
- May include:
  - Sequence number
  - Port numbers
- Does NOT include full route

15 Network Abstraction Truths

- Applications implement application-layer rules
- OS implements transport layer
- Routers modify network-layer headers
- Physical layer frames contain:
  - Port (transport)
  - IP (network)
  - MAC (link)